



2.4.3. Multidimensional Images -

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2.4.3. Multidimensional Images - Practice Assessment

Hyperstacks in ImageJ

0 points possible (ungraded)

Please check all true statements

Check all that apply

☐ Image what you can, not what you need.☒ Hyperstacks can be seen as just stacks with more indexes ✓☐ ImageJ can handle infinite dimensions for stacks☒ In a hyperstack with 10 Z slices and 4 timepoints, adding a single channel means adding 40 images ✓

i Answers are displayed within the problem

Many Dimensions

0 points possible (ungraded)

Open the Mitosis (26MB) dataset from File > Open Samples

This dataset shows a cell undergoing mitosis, from Metaphase to Telophase.

[BioExplorer Article on Mitosis](#)

How many dimensions does this dataset have?

✓ Answer: 5

How many images is this dataset composed of?

✓ Answer: 510

How many Z Slices does this dataset have?

✓ Answer: 5

Navigate the dataset using the sliders. At which timepoint does the DNA of the cells begin to be pulled apart? (Onset of Anaphase)

✓ Answer: 27

i Answers are displayed within the problem

Image Acquisition Considerations

0 points possible (ungraded)

Suppose we are acquiring single channel images of a fluorescent sample in 3D and time. Suppose that each image plane is 2MB (uncompressed) and takes 0.75 sec to acquire.

How much data (in MB) would 20 timepoints with 30 Z slices create?

✓ Answer: 1200

What is the shortest possible time interval between two timepoints for a 10 Z slice stack (in seconds)?

✓ Answer: 7.5

By binning the camera, we can reduce the number of pixels by a certain factor (2,3,4,...). If we bin the camera a factor of 2 (in X and Y), what will be the new size of a single image plane in MB?

✓ Answer: 0.5

i Answers are displayed within the problem

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